

AquaWise — Requirements Specification

2026-09-22 · @Someone

AquaWise is a MERN web portal with three graded functionalities: a household water calculator, a rule-based crisis advisory engine, and a society complaint portal. It maps to UN SDG 6, targets 6.1 (safe drinking water for all) and 6.4 (water-use efficiency).

Item	Value
Course	Web Programming Laboratory, Sem 5, K J Somaiya
Team	2 members
Stack	HTML5, CSS3, Bootstrap 5, JavaScript, React, Node.js/Express, MongoDB, express-session auth, Postman, cloud deploy
Grading	3 working functionalities, code modularity, UI polish, documented testing, presentation

Markers used in this document: **[SRC n]** = number taken from reference n in the References section; **[ASSUMED]** = a design value chosen by us, not from a published source. Every **[ASSUMED]** item is listed again in the Assumptions register at the end.

1. Problem statement

Globally, 2.1 billion people, one in four, still lacked safely managed drinking water in 2024, and 106 million drank untreated surface water [SRC 1]. India's position is acute. NITI Aayog's Composite Water Management Index (June 2018) reports nearly 600 million Indians facing high to extreme water stress and ranks India 120th of 122 countries on water quality [SRC 2]. It also estimates about two lakh deaths a year from inadequate access to safe water, with national demand projected to reach twice the available supply by 2030 [SRC 3].

Urban households add to this pressure without seeing it. Few know how their use compares with the 135–150 litres per capita per day that CPHEEO norms plan for [SRC 4]. Leaks alone can waste over 9,300 gallons (about 35,000 L) per household a year [SRC 8]. When supply fails or water turns unsafe, residents lack clear first steps and a channel to report it. AquaWise closes these gaps with a usage calculator, a crisis advisory engine and a society complaint portal.

Flag: the leak figure [SRC 8] is US household data, used as an order-of-magnitude illustration. The two-lakh-deaths and 2030-demand figures come from a news report of the CWMI [SRC 3] because the NITI PDF could not be opened; cite the PDF directly if you can access it.

2. Objectives

Each objective has a pass/fail check you can demonstrate in the viva and record in the test log. Targets marked [ASSUMED] are our own design targets.

#	Objective	Measure of success
O1	Estimate household daily water use by activity	Calculator output matches hand-computed values from Section 7 formulas within ± 1 L for 10 documented test households
O2	Classify per-capita use against published norms	100% of boundary test cases (19, 20, 54, 55, 99, 100, selected norm, norm+1, 200, 201 lpcd) land in the correct band from Section 7.3
O3	Give correct first-response guidance for 5 water problems	All 5 problem types $\times$ every severity in Section 8 (14 rules) return the right ordered steps and contacts; verified by one Postman/Jest test per rule
O4	Route every complaint to the right handler	100% of complaints filed in testing get the handler mapped to their category (Section 5.3, FR-S12); dashboard counts equal database counts
O5	Deliver a usable, secure, deployed portal	Deployed on a public URL; all protected routes return 401 without a session; Lighthouse accessibility score ≥ 90 on 4 key pages [ASSUMED target]

3. Scope

The build covers the three graded functionalities plus the auth and deployment they depend on; anything needing hardware, payments or live government systems is out.

In scope	Explicitly out of scope
Calculator for one household, with activity breakdown and norm comparison	Metering or IoT sensor readings; any real measured consumption
Saving calculator runs for logged-in users and showing their history	Multi-year analytics, forecasting or ML-based predictions
Rule-based advisory for 5 problem types, 14 rules (Section 8)	AI/chatbot advice; medical diagnosis; lab-grade water-quality testing
Static contact directory: society contacts + national emergency 112 + one municipal example (Mumbai BMC 1916)	A verified contact database for every Indian city; automatic calls to authorities
Signup, login, logout with server-side sessions	OAuth / Google sign-in, OTP, JWT, password-reset email
Create society (admin), join society by code (resident)	Society approval workflows, maintenance billing, visitor management
File complaint, auto-route by category, status updates, dashboard	SMS/WhatsApp/push notifications; file or photo uploads [ASSUMED out, to keep scope to 2 people]
Responsive UI (Bootstrap 5), form validation client + server	Native mobile apps; offline mode; multiple languages
Postman collection + test log; one cloud deployment	Load testing at scale; CI/CD pipelines; custom domain

4. Actors and permissions

Four actors: anyone can use the calculator and advisory; only society members touch complaints, and only the society admin changes their status. Enforce every "No" on the server (middleware), not just by hiding buttons.

Capability	Guest	Resident	Society Admin	System Admin
Use calculator, view result	Yes	Yes	Yes	Yes
Save calculator run, view own history	No	Yes	Yes	Yes

Capability	Guest	Resident	Society Admin	System Admin
Use advisory engine	Yes	Yes	Yes	Yes
Sign up / log in	Yes	—	—	—
Create a society (gets a join code)	No	Yes (becomes its admin)	—	Yes
Join a society by code	No	Yes (one society at a time)	No	No
File a complaint	No	Yes, own society	Yes, own society	No
View complaints	No	Own complaints + society summary counts	All in own society	All
Change complaint status / add resolution note	No	No (may reopen own resolved complaint within 7 days)	Yes, own society	Yes
Edit society details and local contacts	No	No	Yes, own society	Yes
Deactivate users or societies	No	No	Remove member from own society	Yes

- **Guest:** not logged in.
- **Resident:** logged-in user, `role: "resident"`.
- **Society Admin:** resident who created the society, `role: "society_admin"`; one admin per society.
- **System Admin:** seeded account for the team/demo, `role: "system_admin"`; not creatable through signup.

[ASSUMED] One society per user, one admin per society, and the 7-day reopen window are our design choices to keep the model simple.

5. Functional requirements

41 requirements in four groups; the IDs are what your test cases and Postman requests should reference.

5.1 Water Requirement & Usage Calculator (FR-C)

ID	Requirement
FR-C1	The system shall accept household inputs: adults, children, city type, bathing mode, showers per person per day, shower minutes, shower type, bucket baths per person per day, toilet type, washing-machine loads per week, machine type, garden area (m ²), tap-running minutes per person per day, tap type, dripping taps, and optional litres actually received per day.
FR-C2	The system shall pre-fill every optional input with the default in Section 7.2 and show the source beside it.
FR-C3	The system shall compute daily litres for each activity using exactly the formulas in Section 7.2.
FR-C4	The system shall compute household total (L/day) and per-capita use (lpcd) = total ÷ (adults + children).
FR-C5	The system shall display the activity split as a bar or doughnut chart and as a table with litres and % of total.
FR-C6	The system shall classify per-capita use into one band from Section 7.3 and show the band name, colour and the norm it was compared with.
FR-C7	The system shall flag over-use when per-capita use exceeds the norm for the selected city type.
FR-C8	The system shall flag under-supply when litres received per capita is below 55 lpcd, or below the household's estimated need.
FR-C9	The system shall list the top 3 activities by litres with one saving tip each, and the litres saved if the efficient fixture value were used.
FR-C10	The system shall compute results on the client for guests and on the server (POST /api/calculator) for logged-in users, using one shared formula module.

ID	Requirement
FR-C11	The system shall save each logged-in run as a CalculatorLog and show the user's last 10 runs with date, total and band.
FR-C12	The system shall reject invalid inputs with field-level messages per the ranges in Section 9.4.

5.2 Crisis Advisory Engine (FR-A)

ID	Requirement
FR-A1	The system shall let the user pick one of 5 problem types: leakage, no supply, unsafe drinking water, low pressure, contamination.
FR-A2	The system shall ask 1–3 yes/no or choice questions per problem type (Section 8) and derive severity from the answers.
FR-A3	The system shall return the matching rule's ordered steps, numbered, in the order stored.
FR-A4	The system shall show the rule's contacts with name, number and when to call; for the user's own society, contacts come from the Society record.
FR-A5	The system shall show a red emergency banner with 112 for any rule of severity "critical".
FR-A6	The system shall store rules as data (JSON file or MongoDB collection), not hard-coded if/else in UI components.
FR-A7	The system shall expose GET /api/advisory/rules and POST /api/advisory/evaluate (problemType + answers → rule).
FR-A8	The system shall offer a "Report this to my society" button for logged-in members that opens the complaint form pre-filled with category and severity.
FR-A9	The system shall let the user print or copy the steps.

5.3 Society Portal (FR-S)

ID	Requirement
FR-S1	The system shall let a guest sign up with name, email, phone, password.
FR-S2	The system shall log a user in with email + password and create a server-side session.
FR-S3	The system shall log out by destroying the session and clearing the cookie.
FR-S4	The system shall let a resident create a society; the creator becomes its society admin.
FR-S5	The system shall generate a unique 6-character join code for each society.
FR-S6	The system shall let a resident with no society join one by entering a valid code, plus flat number and wing.
FR-S7	The system shall let the society admin regenerate the join code; old codes stop working.
FR-S8	The system shall let the society admin maintain local contacts (role, name, phone, categories handled).
FR-S9	The system shall let a member file a complaint with category, title, description, location in the building and severity.
FR-S10	The system shall assign each new complaint a readable ID (e.g., AQW-2026-0001).
FR-S11	The system shall set new complaints to status "active".
FR-S12	The system shall route each complaint by category to the society contact whose categories include it; if none, to the society admin.
FR-S13	The system shall let the society admin move status active → in_progress → resolved, with a note on each change.

ID	Requirement
FR-S14	The system shall record each status change (who, when, from, to, note) in the complaint's history.
FR-S15	The system shall let the filer reopen a resolved complaint within 7 days, setting it back to active.
FR-S16	The system shall show a society dashboard: counts of active, in-progress and resolved; list filterable by status and category; local contacts panel.
FR-S17	The system shall show residents only their own complaints in detail and society-wide counts.

5.4 Common (FR-G)

ID	Requirement
FR-G1	The system shall provide a landing page explaining SDG 6 and linking to the three features.
FR-G2	The system shall provide a navbar that changes with login state and role.
FR-G3	The system shall return JSON errors as { error: { code, message, fields? } } with correct HTTP status.

6. Non-functional requirements

All numeric targets in this section are [ASSUMED] design targets sized for a 2-person lab project on a free cloud tier; no external source sets them.

ID	Area	Requirement
NFR-U1	Usability	Every page works at 360 px width with no horizontal scroll (Bootstrap grid, tested in Chrome DevTools device mode).
NFR-U2	Usability	A first-time user completes a calculator run in ≤ 2 minutes and an advisory lookup in ≤ 3 clicks.
NFR-U3	Usability	Every input has a visible label, unit (L, m ² , min) and help text; colour is never the only signal for a band (add text + icon).

ID	Area	Requirement
NFR-U4	Usability	Lighthouse accessibility ≥ 90 on Home, Calculator, Advisory, Dashboard.
NFR-V1	Validation	Validate on the client (HTML5 attributes + React state) and again on the server (express-validator or Joi); the server is the authority.
NFR-V2	Validation	Numeric inputs are integers or decimals inside the ranges in Section 9.4; out-of-range returns HTTP 400 with the field name.
NFR-V3	Validation	Strings are trimmed; free text is length-capped and HTML-escaped on output (React escapes by default; never use dangerouslySetInnerHTML for user text).
NFR-S1	Security	Passwords hashed with bcrypt, cost factor 12; never returned by any API.
NFR-S2	Security	Sessions via express-session + connect-mongo; cookie httpOnly, sameSite "lax", secure in production, maxAge 2 h; session ID regenerated on login.
NFR-S3	Security	Role/ownership checks in middleware (requireAuth, requireRole, requireSameSociety) on every protected route.
NFR-S4	Security	Login rate-limited to 5 failed attempts per 15 min per IP (express-rate-limit).
NFR-S5	Security	Secrets (Mongo URI, session secret) only in environment variables; .env in .gitignore.
NFR-S6	Security	helmet() enabled; CORS restricted to the deployed frontend origin with credentials: true.
NFR-S7	Security	Queries built with Mongoose only; reject keys beginning with \$ in request bodies (express-mongo-sanitize) to block NoSQL injection.
NFR-P1	Performance	API p95 response ≤ 500 ms for non-aggregate routes on the deployed instance with ≤ 20 concurrent users.
NFR-P2	Performance	Calculator result renders ≤ 100 ms after submit for guests (client-side computation).

ID	Area	Requirement
NFR-P3	Performance	Indexes on User.email, Society.joinCode, Complaint {society, status}, CalculatorLog {user, createdAt}.
NFR-P4	Performance	Dashboard complaint list paginated, 20 per page.
NFR-M1	Modularity	Backend split into routes / controllers / models / middleware / services (calculator + advisory as pure-function services); frontend into pages / components / api / hooks.
NFR-M2	Modularity	The calculator formula module and advisory rules are unit-tested independently of Express and React.

7. Water-usage formulas and benchmarks

Daily household use is the sum of eight activity terms; per-capita use is that total divided by household size, then compared with Indian (CPHEEO, BIS, JJM) and WHO norms. US-unit sources are converted at 1 US gallon = 3.785411784 L (exact by definition); values below are rounded to 2 decimals.

7.1 Per-capita benchmarks

Benchmark	Value (lpcd)	Meaning	Source
WHO — no access	< 5	Consumption not assured; very high health concern	[SRC 5]
WHO — basic access	≤ 20	Drinking + basic hygiene only; high health concern	[SRC 5]
WHO — intermediate access	~50	Personal, food hygiene, laundry, bathing assured; low concern	[SRC 5]
Jal Jeevan Mission service level	55	Minimum "adequate" supply per person (rural tap connections), quality per BIS 10500	[SRC 6]
CPHEEO — towns with piped water, no sewerage	70	Planning norm	[SRC 4]
WHO — optimal access	≥ 100	All consumption and hygiene needs met; very low concern	[SRC 5]

Benchmark	Value (lpcd)	Meaning	Source
CPHEEO — cities with piped water + sewerage	135	Planning norm	[SRC 4]
CPHEEO — metropolitan/mega cities with sewerage	150	Planning norm	[SRC 4]
BIS IS 1172:1993 — bigger cities, full flushing	150–200	Upper design range	[SRC 4]
WHO — drinking + food preparation minimum	7.5	Meets most people's needs under most conditions	[SRC 5]

Mumbai users select "Metropolitan" (N = 150). The user's city-type choice sets N = 70, 135 or 150.

7.2 Activity formulas

Let A = adults, C = children, P = A + C.

$$L_{\text{total}} = L_{\text{bath}} + L_{\text{toilet}} + L_{\text{laundry}} + L_{\text{garden}} + L_{\text{tap}} + L_{\text{drink\&cook}} + L_{\text{utensils\&clean}} + L_{\text{leak}}$$

$$L_{\text{bath}} = P \cdot (s \cdot m \cdot q_{\text{shower}} + b \cdot V_{\text{bucket}})$$

$$L_{\text{toilet}} = P \cdot f \cdot V_{\text{flush}} \quad L_{\text{laundry}} = \frac{w \cdot V_{\text{load}}}{7} \quad L_{\text{garden}} = G \cdot 3.63$$

$$L_{\text{tap}} = P \cdot t \cdot q_{\text{tap}} \quad L_{\text{drink\&cook}} = 7.5 \cdot P \quad L_{\text{utensils\&clean}} = 20 \cdot P \quad L_{\text{leak}} = 31.11 \cdot d$$

$$\text{lpcd} = \frac{L_{\text{total}}}{P}$$

Symbol	Input / constant	Default or options (litres)	Source
s	Showers per person per day	1 [ASSUMED default]	User input
m	Minutes per shower	7.8 min	[SRC 12] REU2016 average

Symbol	Input / constant	Default or options (litres)	Source
q_shower	Shower flow	Standard 9.46 L/min (2.5 gpm); Efficient 7.57 L/min (2.0 gpm)	[SRC 7]
b	Bucket baths per person per day (if bathing mode = bucket)	0	User input
V_bucket	Litres per bucket bath	20 L	[ASSUMED] — ask users to measure their bucket
f	Flushes per person per day	5.0	[SRC 12]
V_flush	Litres per flush	Old 22.71 (6 gpf); Standard 6.06 (1.6 gpf); Efficient 4.85 (1.28 gpf)	[SRC 10]
w	Washing-machine loads per week	User input	—
V_load	Litres per load	Older/top-load avg 117.35 (31 gal); Standard 75.71 (20 gal); Efficient 53.00 (14 gal)	[SRC 12], [SRC 11]
G	Garden area watered, m ²	0	User input
3.63	L per m ² per day	1 inch (25.4 mm = 25.4 L/m ²) per week ÷ 7, rainfall included	[SRC 13]
t	Minutes tap left running per person per day (brushing, shaving, dishes under running tap)	2 min [ASSUMED default]	User input
q_tap	Tap flow	Standard 8.33 L/min (2.2 gpm); Aerated 5.68 L/min (1.5 gpm)	[SRC 9]
7.5	Drinking + cooking, L/person/day	Fixed	[SRC 5]

Symbol	Input / constant	Default or options (litres)	Source
20	Utensils (10) + house cleaning (10), L/person/day	Fixed	[ASSUMED] — widely taught IS 1172 breakup; not verified against the standard itself
d	Taps dripping ~1 drop/second	0	User input
31.11	L/day per dripping tap	3,000 gal/year ÷ 365	[SRC 8]

[ASSUMED] Children use the same per-person constants as adults; no authoritative child-specific litre figure was found. A and C are still stored separately for display and future tuning.

7.3 Band classification (evaluate top to bottom, first match wins)

Order	Condition ($x = \text{Ipcd}$, $N = \text{selected city norm}$)	Band	Flag	Basis
1	$x < 20$	Critical under-supply	Under-supply	Below WHO basic [SRC 5]
2	$x < 55$	Under-supply	Under-supply	Below JJM 55 [SRC 6] and WHO intermediate ~50 [SRC 5]
3	$x > 200$	Severe over-use	Over-use	Above IS 1172 upper 200 [SRC 4]
4	$x > N$	Over-use	Over-use	Above CPHEEO norm [SRC 4]
5	$x < 100$	Basic	None	Meets 55, below WHO optimal 100 [SRC 5]
6	otherwise	Within norm	None	$100 \leq x \leq N$

Supply check (FR-C8), only when the user enters litres received per day R : supplied $\text{Ipcd} = R \div P$. If $R \div P < 55 \rightarrow$ flag "Under-supply vs JJM norm" [SRC 6]. Else if $R < L_{\text{total}} \rightarrow$ flag "Supply below your estimated need by $(L_{\text{total}} - R)$ L/day".

7.4 Worked example (use as test case TC-C01)

2 adults + 2 children, Metropolitan (N = 150), 1 standard shower/person/day of 7.8 min, standard toilet, 5 standard machine loads/week, no garden, 2 min standard tap/person/day, 1 dripping tap.

Activity	Calculation	L/day
Bathing	$4 \times 1 \times 7.8 \times 9.46$	295.15
Toilet	$4 \times 5 \times 6.06$	121.20
Laundry	$5 \times 75.71 \div 7$	54.08
Garden	0×3.63	0.00
Taps	$4 \times 2 \times 8.33$	66.64
Drinking + cooking	4×7.5	30.00
Utensils + cleaning	4×20	80.00
Leak	1×31.11	31.11
Total		678.18

Per capita = $678.18 \div 4 = 169.55$ lpcd $\rightarrow$ rule 4, **Over-use** (above 150). Switching to efficient shower, toilet and machine drops it to 578.79 L/day, 144.70 lpcd, inside the norm; show that as the FR-C9 saving tip.

Flag: most fixture figures are US (EPA/ENERGY STAR/WRF) because no Indian open-data source gives per-fixture litres. They are reasonable for Indian urban flats with Western fixtures but are not India-measured values; say so in the UI footnote.

8. Advisory rule table

14 rules across 5 problem types; the engine asks the trigger questions, picks the first matching rule from the most severe down, and returns its steps in order. Severity levels: **low** (inconvenience), **medium** (fix within days), **high** (health or property risk), **critical** (act now; shows the 112 banner).

flowchart LR

```

A[Pick problem type] --> B[Answer 1-3 questions]
B --> C{Match rules<br/>critical first}
C --> D[Ordered steps]
D --> E[Contacts]
E --> F[Report to society<br/>pre-filled complaint]
```

The complaint pre-fill maps problem type to Complaint.category one-to-one.

Contact keys used below: PLUMBER = licensed plumber; SOC_MAINT = society maintenance in-charge (from Society.contacts); SOC_ADMIN = society admin/secretary; MUNI = municipal water department (Mumbai: BMC helpline 1916 [SRC 15]; other cities: configurable); DOCTOR = nearest doctor/PHC; ERSS = 112, India's single emergency number routing to police, health and fire [SRC 16].

Rule	Problem	Severity	Trigger (user answers)	Ordered steps	Contacts
LK-1	Leakage	low	Dripping tap or running flush inside own flat; no wet walls	1) Put a container under the drip. 2) Close the small valve under that tap or cistern if there is one. 3) Replace the washer/cartridge or flush flapper, or book a plumber. 4) Enter it as a dripping tap in the calculator: each one wastes about 31 L/day [SRC 8].	PLUMBER, SOC_MAINT
LK-2	Leakage	medium	Pipe-joint leak, wall seepage or tank overflow; no standing water	1) Close your flat's main inlet valve. 2) Unplug appliances and keep sockets near the wet wall off. 3) Photograph and mark the spot. 4) File a complaint (leakage). 5) Reopen the valve only for short periods until repaired.	SOC_MAINT, PLUMBER
LK-3	Leakage	critical	Burst pipe, water flooding the floor, or water near switches, meters or the lift	1) Move everyone away from the water. 2) Switch off the flat's main breaker only if you can reach it without standing in water. 3) Close the main inlet valve; if it is a common line, ask security to close the building valve. 4) Call SOC_ADMIN; call 112 for electrical hazard or injury. 5) If the burst is on the road main, report to MUNI. 6) File a complaint.	ERSS, SOC_ADMIN, MUNI

Rule	Problem	Severity	Trigger (user answers)	Ordered steps	Contacts
NS-1	No supply	low	Outage under 24 h, or announced cut; stored water available	1) Check the society notice board and municipal announcements for a scheduled cut. 2) Ask neighbours: is it only your flat or the building? 3) Only your flat: check that your inlet valve is open. 4) Save stored water for drinking and cooking first (7.5 L/person/day minimum [SRC 5]).	SOC_ADMIN
NS-2	No supply	medium	Unannounced outage over 24 h across the building	1) Ask SOC_ADMIN to check tank levels and the pump. 2) SOC_ADMIN registers a complaint with MUNI and shares the complaint number. 3) Ration to at least 20 L/person/day for drinking and basic hygiene [SRC 5]. 4) File an AquaWise complaint (no supply).	SOC_ADMIN, MUNI
NS-3	No supply	critical	Outage over 72 h, or under 7.5 L per person left, or an infant, elderly or ill member at home	1) Ask SOC_ADMIN to arrange a municipal or authorised tanker. 2) Treat tanker water before drinking as in UD-1 step 3. 3) Use water only for drinking and cooking. 4) SOC_ADMIN escalates in writing to the ward office, quoting the complaint number. 5) Call 112 if anyone shows signs of dehydration.	SOC_ADMIN, MUNI, ERSS

Rule	Problem	Severity	Trigger (user answers)	Ordered steps	Contacts
UD-1	Unsafe drinking water	high	Odd taste, smell or colour, or tank not cleaned in over 6 months; nobody ill	1) Stop drinking untreated tap water; use sealed bottled or treated water. 2) If cloudy, filter through a clean cloth or let it settle and draw off the clear water [SRC 14]. 3) Boil at a rolling boil for 1 minute, or add 2 drops of unscented 5–9% bleach per litre (8 drops per US gallon) and wait 30 minutes; double the bleach if cloudy [SRC 14]. 4) Store in clean, covered containers [SRC 14]. 5) Ask SOC_ADMIN to clean the tanks and test water against BIS IS 10500 [SRC 6].	SOC_ADMIN, MUNI
UD-2	Unsafe drinking water	critical	Anyone at home has diarrhoea, vomiting or fever after drinking the water	1) See a doctor; call 112 if symptoms are severe or the patient is an infant or elderly. 2) Stop using that source entirely. 3) Treat all drinking water as in UD-1 step 3. 4) Tell SOC_ADMIN so other residents are warned. 5) SOC_ADMIN reports to MUNI.	DOCTOR, ERSS, SOC_ADMIN, MUNI
LP-1	Low pressure	low	Only one tap or shower is weak	1) Unscrew and clean the aerator or shower-head mesh. 2) Check the valve under that fixture is fully open. 3) If still weak, book a plumber.	PLUMBER
LP-2	Low pressure	medium	Whole flat is weak; neighbours are fine	1) Check your flat's main inlet valve is fully open. 2) Leak test: close every tap and watch the meter; if it moves, follow LK-2. 3) Ask a plumber to check the inlet filter and pipes for blockage. 4) File a complaint (low pressure).	SOC_MAINT, PLUMBER

Rule	Problem	Severity	Trigger (user answers)	Ordered steps	Contacts
LP-3	Low pressure	medium	Whole building or area is weak	1) Ask SOC_ADMIN to check the overhead tank level and pump. 2) Check municipal supply timings for your ward. 3) SOC_ADMIN registers a complaint with MUNI. 4) Store water during supply hours.	SOC_ADMIN, MUNI
CT-1	Contamination	high	Brown/yellow water or sediment; no sewage smell	1) Do not drink it. 2) Run the cold tap until it clears; collect a sample in a clean bottle. 3) Treat drinking water as in UD-1 steps 2–4. 4) File a complaint (contamination). 5) Society cleans the tank; test against IS 10500 [SRC 6].	SOC_ADMIN, SOC_MAINT, MUNI
CT-2	Contamination	critical	Sewage smell, black water, worms or visible matter, or known sewage-line mixing	1) Do not drink, cook or brush teeth with it; use sealed bottled water. 2) Photograph and collect a sample. 3) SOC_ADMIN isolates the affected tank and reports to MUNI immediately. 4) SOC_ADMIN warns all residents through the dashboard. 5) Reuse only after tank cleaning, disinfection and a passing IS 10500 test [SRC 6]. 6) Anyone ill: follow UD-2.	MUNI, SOC_ADMIN, ERSS
CT-3	Contamination	critical	Chemical, fuel or oily smell, or an oily film	1) Do not drink or cook with it; boiling will not remove chemicals. 2) Use sealed bottled water. 3) Photograph and collect a sample. 4) SOC_ADMIN reports to MUNI immediately and warns residents. 5) Anyone ill: call 112.	MUNI, SOC_ADMIN, ERSS

[ASSUMED] The time thresholds (24 h, 72 h, 6 months) and the severity labels are our design choices, not from a standard. "Boiling will not remove chemicals" (CT-3) and the physical safety steps (breaker, valves) are general safety practice with no source opened here. The 2-drops-per-litre figure is our conversion of CDC's 8 drops per gallon. Replace 1916 with the correct municipal number for other cities.

9. Data model (MongoDB / Mongoose)

Four collections; Society embeds its contacts and Complaint embeds its status history, so each dashboard view is one query. Every collection gets Mongoose `timestamps: true` (`createdAt`, `updatedAt`: Date, auto). All length limits and ranges are [ASSUMED] design values.

```
erDiagram
    USER }o--o| SOCIETY : "member of"
    SOCIETY ||--|| USER : "admin"
    SOCIETY ||--o{ COMPLAINT : has
    USER ||--o{ COMPLAINT : files
    USER ||--o{ CALCULATORLOG : saves
```

9.1 User

Field	Type	Required	Validation / rules
_id	ObjectId	auto	—
name	String	yes	trim; 2–60 chars; letters, spaces, . ' - only
email	String	yes	trim, lowercase; valid email format; unique index
phone	String	yes	exactly 10 digits, first digit 6–9 (Indian mobile)
passwordHash	String	yes	bcrypt hash (cost 12); <code>select: false</code> ; plain password rule: 8–64 chars, ≥ 1 letter and ≥ 1 digit
role	String enum	yes	resident society_admin system_admin; default resident; never settable from signup body
society	ObjectId → Society	no	null until joined; one society per user

Field	Type	Required	Validation / rules
flatNo	String	if society set	trim; 1-10 chars; A-Z, 0-9, - /
wing	String	no	trim; 0-10 chars
isActive	Boolean	yes	default true; inactive users cannot log in
lastLoginAt	Date	no	set on successful login

9.2 Society

Field	Type	Required	Validation / rules
_id	ObjectId	auto	—
name	String	yes	trim; 3-80 chars
joinCode	String	yes	exactly 6 chars from A-Z and 2-9 (no 0, O, 1, l); uppercase; unique index ; regenerated on request (FR-S7)
address.line1	String	yes	5-120 chars
address.area	String	yes	2-60 chars
address.city	String	yes	2-60 chars; default "Mumbai"
address.pincode	String	yes	exactly 6 digits, first digit 1-9
cityType	String enum	yes	metro city_sewered town_unsewered; sets norm N = 150 / 135 / 70 [SRC 4]
municipalHelpline	String	no	3-15 digits; default "1916" when city is Mumbai [SRC 15]
admin	ObjectId → User	yes	must be a member; exactly one
memberCount	Number	yes	integer ≥ 1; updated on join/leave
contacts	Array of Contact	no	max 20 entries
contacts[].role	String	yes	2-40 chars, e.g., "Maintenance in-charge", "Plumber"
contacts[].name	String	yes	2-60 chars

Field	Type	Required	Validation / rules
contacts[].phone	String	yes	10 digits (mobile) or 3–15 digits (landline/helpline)
contacts[].categories	[String enum]	yes	≥ 1 of the Complaint.category values; used for routing (FR-S12)
isActive	Boolean	yes	default true

9.3 Complaint

Field	Type	Required	Validation / rules
_id	ObjectId	auto	—
complaintNo	String	yes	format AQW-YYYY-NNNN, sequential per year; unique index
society	ObjectId → Society	yes	must equal the filer's society
filedBy	ObjectId → User	yes	from session, never from body
category	String enum	yes	leakage no_supply unsafe_drinking_water low_pressure contamination other
severity	String enum	yes	low medium high critical; pre-filled from advisory rule, editable by filer
advisoryRuleId	String	no	one of LK-1...CT-3 when filed from the advisory (FR-A8)
title	String	yes	trim; 5–80 chars
description	String	yes	trim; 10–1000 chars
location	String enum	yes	own_flat common_area overhead_tank underground_tank road_main

Field	Type	Required	Validation / rules
assignedContact	Object {role, name, phone}	yes	snapshot copied at filing: first Society.contacts entry whose categories include this category; else the admin's name and phone
status	String enum	yes	active in_progress resolved; default active
statusHistory	Array	yes	appended on every change; never edited
statusHistory[.].from	String enum null	yes	null for the first entry
statusHistory[.].to	String enum	yes	allowed moves only: active→in_progress, in_progress→resolved, active→resolved, resolved→active (reopen, filer only, ≤ 7 days)
statusHistory[.].by	ObjectId → User	yes	from session
statusHistory[.].note	String	yes for admin moves	0–300 chars
statusHistory[.].at	Date	yes	server time
resolvedAt	Date	no	set when status becomes resolved; cleared on reopen
Indexes	—	—	{ society: 1, status: 1, createdAt: -1 }; { filedBy: 1, createdAt: -1 }

9.4 CalculatorLog

Field	Type	Required	Validation / rules
_id	ObjectId	auto	—
user	ObjectId → User	yes	from session
inputs.adults	Number	yes	integer 1–20

Field	Type	Required	Validation / rules
inputs.children	Number	yes	integer 0–20; adults + children $\leq$ 30
inputs.cityType	String enum	yes	metro city_sewered town_unsewered
inputs.bathingMode	String enum	yes	shower bucket both
inputs.showersPerPersonPerDay	Number	if mode $\neq$ bucket	0–5, step 0.5
inputs.showerMinutes	Number	if mode $\neq$ bucket	1–60; default 7.8 [SRC 12]
inputs.showerType	String enum	if mode $\neq$ bucket	standard efficient
inputs.bucketBathsPerPersonPerDay	Number	if mode $\neq$ shower	0–5, step 0.5
inputs.toiletType	String enum	yes	old standard efficient
inputs.machineLoadsPerWeek	Number	yes	integer 0–21
inputs.machineType	String enum	if loads > 0	older standard efficient
inputs.gardenAreaM2	Number	yes	0–1000; default 0
inputs.tapMinutesPerPersonPerDay	Number	yes	0–60; default 2
inputs.tapType	String enum	yes	standard aerated
inputs.drippingTaps	Number	yes	integer 0–20; default 0
inputs.suppliedLitresPerDay	Number	no	0–20000
results.byActivity	Object of Numbers	yes	keys bath, toilet, laundry, garden, tap, drinkCook, utensilsClean, leak; each $\geq$ 0, 2 decimals
results.totalLitresPerDay	Number	yes	$\geq$ 0; equals sum of byActivity $\pm$ 0.01

Field	Type	Required	Validation / rules
results.perCapitaLpcd	Number	yes	total ÷ (adults + children)
results.normLpcd	Number	yes	70 135 150
results.band	String enum	yes	critical_undersupply undersupply basic within_norm overuse severe_overuse
results.flags	[String enum]	yes	overuse undersupply_norm supply_below_need; may be empty
formulaVersion	String	yes	e.g., "1.0"; bump when constants change so old logs stay explainable
Index	—	—	{ user: 1, createdAt: -1 }

Results are always recomputed on the server; any `results` sent by the client are ignored.

10. Assumptions register

These are the values we chose ourselves or took from secondary sources; state them on your "Assumptions" slide so examiners see they are deliberate.

#	Assumption	Where used	Why / how to strengthen
A1	Bucket bath = 20 L	7.2 V_bucket	No source found; measure a typical bucket and replace
A2	Utensils 10 + house cleaning 10 L/person/day	7.2	Commonly taught IS 1172 breakup; not checked against the standard text
A3	Children use the same per-person constants as adults	7.2	No child-specific norm found
A4	Defaults: 1 shower/person/day, 2 tap-running min/person/day	7.2	Only defaults; user overrides them

#	Assumption	Where used	Why / how to strengthen
A5	Fixture litres from US sources (EPA, ENERGY STAR, WRF)	7.2	No Indian per-fixture open data found; footnote in UI
A6	Garden 3.63 L/m ² /day includes rainfall	7.2	EPA guidance; overstates need in monsoon
A7	CPHEEO/IS 1172 values read from a secondary summary	7.1	Cite the CPHEEO Manual (1999) and IS 1172:1993 directly if you can get them from the library
A8	CWMI deaths and 2030 demand figures via a news report	1	NITI PDF did not load; cite it directly
A9	Advisory time thresholds (24 h, 72 h, 6 months) and severity labels	8	Design choice
A10	Municipal helpline defaults to BMC 1916	8, 9.2	Mumbai only; configurable per society
A11	All NFR numbers, field lengths and input ranges	6, 9	Design targets for a lab project
A12	One society per user, one admin per society, 7-day reopen, no photo uploads	3, 4, 9	Scope control for a 2-person team

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All pages were opened on 22 Sep 2026.

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